

14. Modul Komputasi Awan - Deploy Aplikasi berbasis Cluster Container dengan Docker swarm dan Traefik Reverse Proxy - study case Wordpress

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Prasyarat

- Internet
- PC/Laptop Sudah Terinstall VirtualBox
- Sudah ada VM **Container-Manager, Container-Worker1, Container-Worker2**
- Sudah terinstall docker, docker-compose, swarm cluster

Instalasi Aplikasi Wordpress

Wordpress ini terdiri dari aplikasi berbasis web dan database mysql, kita akan buat container terpisah antara web dan databasenya namun masih dalam 1 stack. Skenarionya Aplikasi akan kita replica menjadi 4 container dan akan running di seluruh worker, sedangkan database hanya akan jalan 1 container saja dan akan running di Worker2 , project ini akan kita namakan **wpstack**

1. Remote SSH ke **Container-Manager**, gunakan ip yang sudah dialokasikan diatas yaitu **192.168.56.10**

```
ssh umam@192.168.56.10
```

2. buat directory untuk wpstack

```
mkdir ~/wpstack
```

3. masuk ke directory tersebut

```
cd wpstack
```

4. buat file wordpress.yml

```
nano wordpress.yml
```

4. isi file konfigurasi tersebut seperti dibawah, lalu simpan

```
services:
  wordpress:
    image: wordpress:latest
    environment:
      WORDPRESS_DB_HOST: db
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppass
      WORDPRESS_DB_NAME: wpdb
    networks:
      - traefik-public
  deploy:
    replicas: 4
    placement:
      constraints:
        - node.labels.role == worker
    labels:
      - "traefik.enable=true"
      - "traefik.http.routers.wordpress.rule=Host(`wp.komputasiawan.lab`)"
      - "traefik.http.services.wordpress.loadbalancer.server.port=80"
      - "traefik.http.routers.wordpress.entrypoints=web"

  db:
    image: mysql:8.0
    environment:
```

```

MYSQL_DATABASE: wpdb
MYSQL_USER: wpuser
MYSQL_PASSWORD: wppass
MYSQL_ROOT_PASSWORD: rootpass
volumes:
  - db-data:/var/lib/mysql
networks:
  - traefik-public
deploy:
  placement:
    constraints:
      - node.hostname == Container-Worker2

networks:
  traefik-public:
    external: true

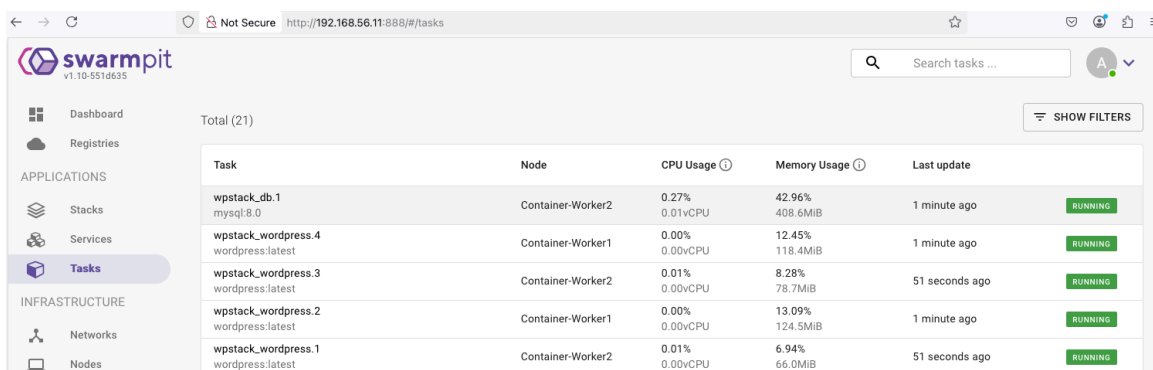
volumes:
  db-data:

```

5. Deploy traefik menggunakan perintah

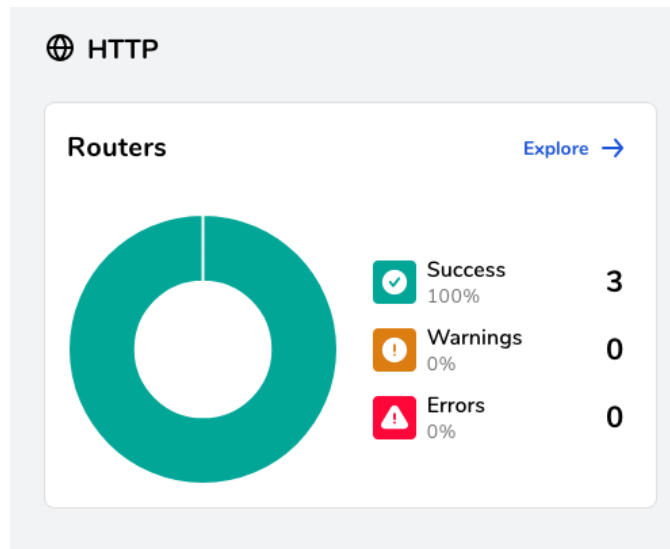
```
sudo docker stack deploy -c wordpress.yml wpstack
```

6. bisa sambil buka swarmpit untuk memantau proses deploy stack tersebut di menu Tasks, pastikan semuanya running



Task	Node	CPU Usage	Memory Usage	Last update	Status
wpstack_db.1 mysql:8.0	Container-Worker2	0.27% 0.01vCPU	42.96% 408.6MiB	1 minute ago	RUNNING
wpstack_wordpress.4 wordpress:latest	Container-Worker1	0.00% 0.00vCPU	12.45% 118.4MiB	1 minute ago	RUNNING
wpstack_wordpress.3 wordpress:latest	Container-Worker2	0.01% 0.00vCPU	8.28% 78.7MiB	51 seconds ago	RUNNING
wpstack_wordpress.2 wordpress:latest	Container-Worker1	0.00% 0.00vCPU	13.09% 124.5MiB	1 minute ago	RUNNING
wpstack_wordpress.1 wordpress:latest	Container-Worker2	0.01% 0.00vCPU	6.94% 66.0MiB	51 seconds ago	RUNNING

7. buka traefik , klik Explore pada http



8. pastikan subdomain wp.komputasiawab.lab sudah muncul

The screenshot shows the Traefik dashboard at `http://192.168.56.10:8080/dashboard/#/http/routers`. The dashboard lists 4 HTTP Routers, 5 HTTP Services, and 2 HTTP Middlewares. Below the summary, a table provides details for each router:

Status	TLS	Rule	Entrypoints	Name
✓		PathPrefix(`/api`)	traefik	api@internal
✓		PathPrefix(`/`)	traefik	dashboard@internal
✓		Host(`coba.komputasiawan.lab`)	web	web@docker
✓		Host(`wp.komputasiawan.lab`)	web	wordpress@docker

5. Pointing domain wp.komputasiawan.lab ke ip 192.168.56.10 dengan langkah-langkah seperti pada modul sebelumnya sehingga menjadi seperti berikut di konfigurasinya

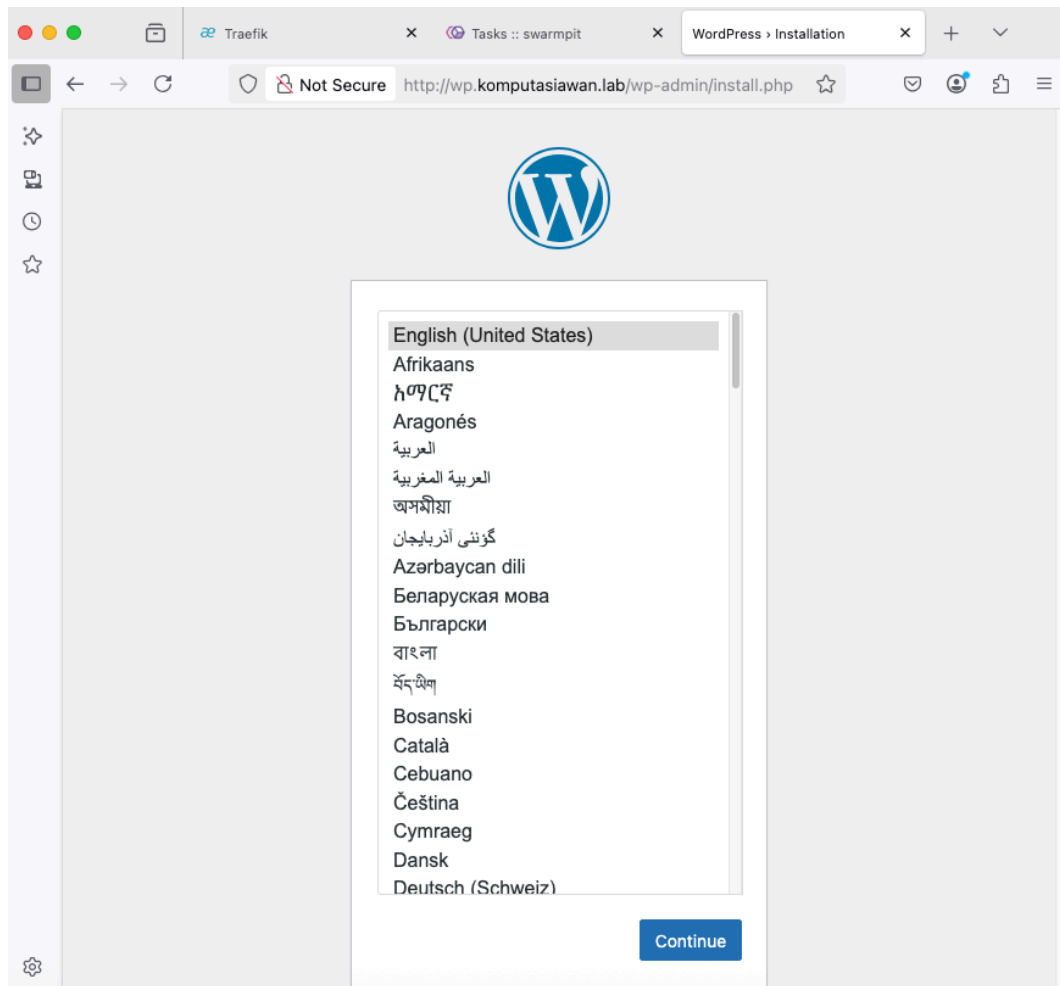
 *hosts - Notepad

```
File Edit Format View Help
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#       102.54.94.97       rhino.acme.com          # source server
#       38.25.63.10       x.acme.com              # x client host

# localhost name resolution is handled within DNS itself.
#       127.0.0.1         localhost
#       ::1               localhost

192.168.56.10    coba.komputasiawan.lab  wp.komputasiawan.lab
```

6. buka <http://wp.komputasiawan.lab> di browser , lalu akan otomatis diarahkan ke halaman proses instalasi wordpress , klik Continue



7. Isi site title , username, password, serta email, sesuaikan dengan data anda, lalu klik install Wordpress



Welcome

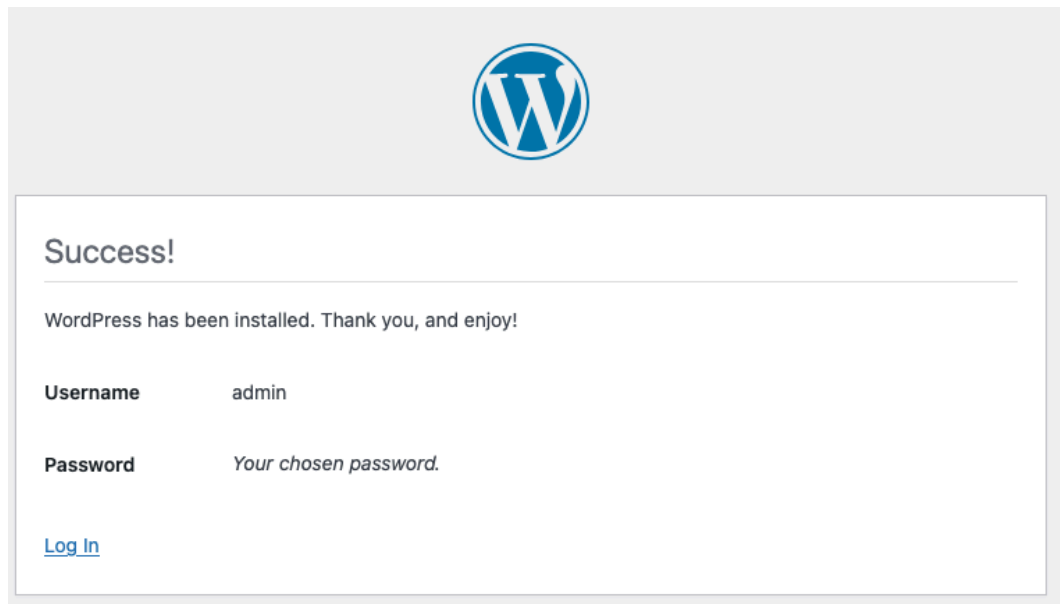
Welcome to the famous five-minute WordPress installation process! Just fill in the information below and you'll be on your way to using the most extendable and powerful personal publishing platform in the world.

Information needed

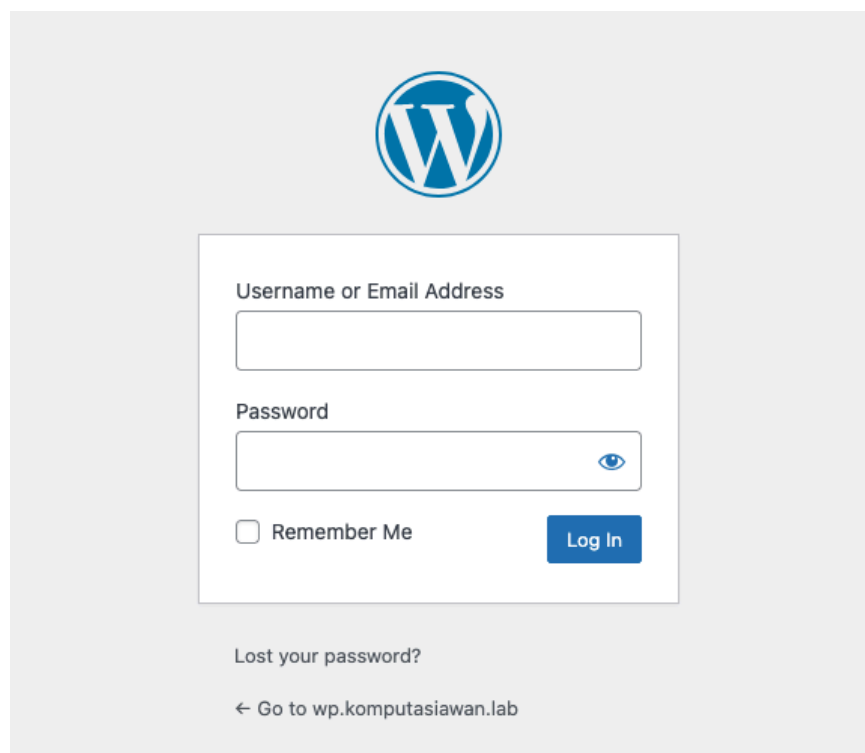
Please provide the following information. Do not worry, you can always change these settings later.

Site Title	wp.komputasiawan.lab
Username	admin Usernames can have only alphanumeric characters, spaces, underscores, hyphens, periods, and the @ symbol.
Password	Udinus2025jaya Strong Important: You will need this password to log in. Please store it in a secure location.
Your Email	chaerul@dsn.dinus.ac.id Double-check your email address before continuing.
Search engine visibility	☐ Discourage search engines from indexing this site It is up to search engines to honor this request.
Install WordPress	

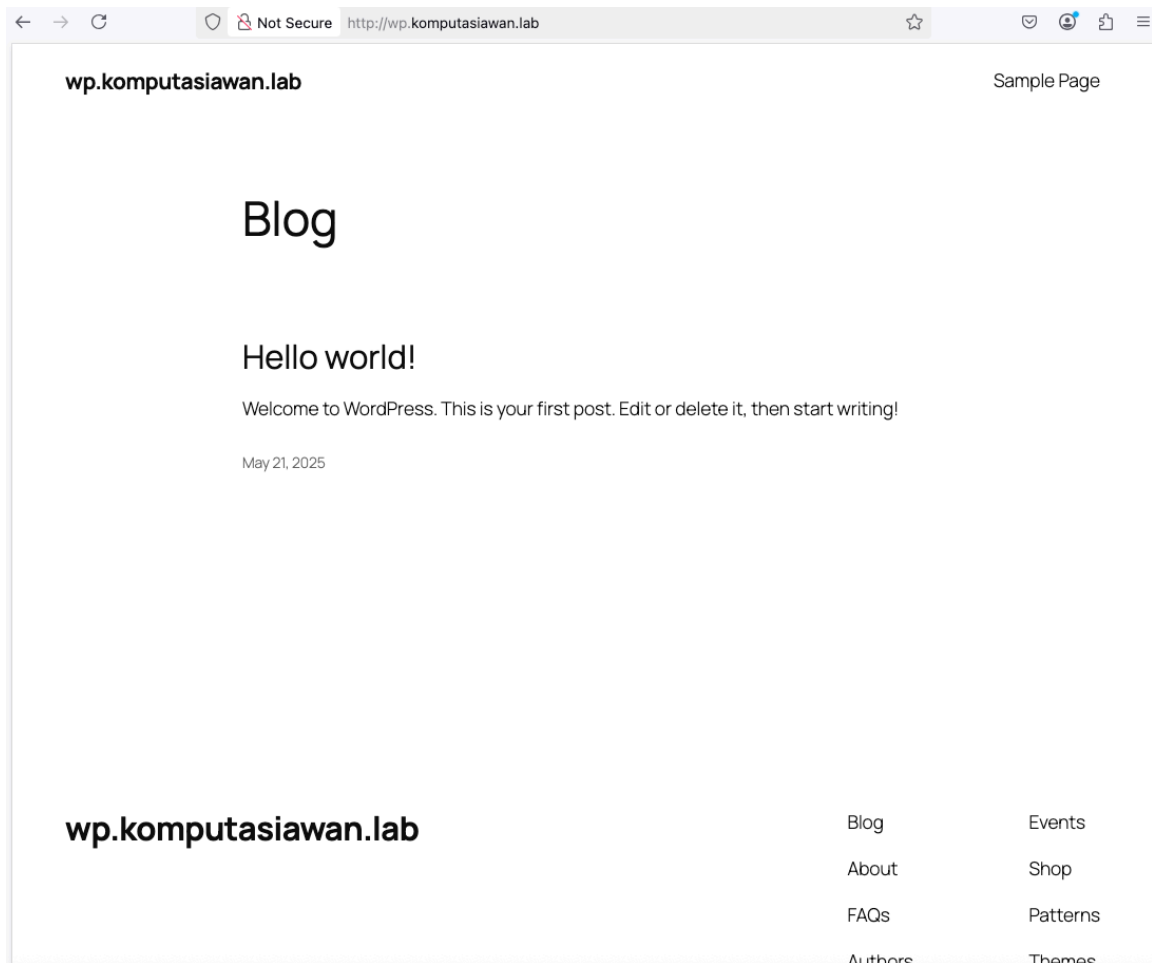
8. Jika sudah sampai halaman konfirmasi sukses artinya sudah berhasil menginstall wordpress, klik login



9. Ini adalah halaman untuk login ke wordpressnya, klik langsung saja Go to wp.komputasiawan.lab untuk langsung ke halaman utamanya



10. Ini adalah halaman website hasil installasi wordpress barusan



16. Silahkan sambil dilihat dashboardnya swarmpit, semakin banyak aktifitas yang kita lakukan maka dashboardnya akan makin CANTIK

